AFRL-AFOSR-VA-TR-2025-0186

Novel Collisional Particle-In-Cell (CPIC) Methods for Kinetic Equations in Plasma Physics

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03/03/2025
Final Technical Report

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Air Force Research Laboratory
Air Force Office of Scientific Research
Arlington, Virginia 22203
Air Force Materiel Command

REPORT DOCUMENTATION PAGE

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1. REPORT DATE	**2. REPORT TYPE**	**3. DATES COVERED** START DATE	END DATE
20250303	Final	20210915	20240914

4. TITLE AND SUBTITLE
Novel Collisional Particle-In-Cell (CPIC) Methods for Kinetic Equations in Plasma Physics

5a. CONTRACT NUMBER	**5b. GRANT NUMBER**	**5c. PROGRAM ELEMENT NUMBER**
	FA9550-21-1-0358	61102F
5d. PROJECT NUMBER	**5e. TASK NUMBER**	**5f. WORK UNIT NUMBER**

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8. PERFORMING ORGANIZATION REPORT NUMBER

9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES)	**10. SPONSOR/MONITOR'S ACRONYM(S)**	**11. SPONSOR/MONITOR'S REPORT NUMBER(S)**
Air Force Office of Scientific Research 875 N. Randolph St. Room 3112 Arlington, VA 22203	AFRL/AFOSR RTA2	AFRL-AFOSR-VA-TR-2025-0186

12. DISTRIBUTION/AVAILABILITY STATEMENT
A Distribution Unlimited: PB Public Release

13. SUPPLEMENTARY NOTES

14. ABSTRACT
The overall objective of this project is to develop a novel structure-preserving collisional particle-in-cell (CPIC) method for kinetic plasma simulations. The key idea stems from the PI's recent work on solving the spatially homogeneous Landau equation. By interpreting the Landau operator as a “transport” operator with a properly regularized velocity field, we can define a particle solution. Moreover, the method can preserve critical physical properties, including the conservation of charge, momentum, and energy, as well as the decay of entropy. This new “characteristic” perspective of the Landau equation offers a natural particle framework to handle collisions and field effects in a unified manner and opens door to many exciting research directions. Specifically, we will pursue the following three aims, which, upon integration, will form a complete CPIC method for the full Vlasov-Landau-Maxwell (VLM) system: 1) Establishing proper particle regularization for the spatially dependent Landau collision operator. 2) Developing structure-preserving particle-field solver for the VLM system and validation it against plasma benchmarks. 3) Enhancing the efficiency and accuracy of the proposed CPIC method.

15. SUBJECT TERMS

16. SECURITY CLASSIFICATION OF: a. REPORT	b. ABSTRACT	c. THIS PAGE	**17. LIMITATION OF ABSTRACT**	**18. NUMBER OF PAGES**
U	U	U	UU	4

19a. NAME OF RESPONSIBLE PERSON	**19b. PHONE NUMBER** *(Include area code)*
FARIBA FAHROO	426-8429

Standard Form 298 (Rev.5/2020)
Prescribed by ANSI Std. Z39.18

Technical Report (Final)

Accomplishments

- **Research Objectives: Please list the main research objectives of this project**

  The overall objective of this project is to develop a novel structure-preserving collisional particle-in-cell (CPIC) method for kinetic plasma simulations. The key idea stems from the PI's recent work on solving the spatially homogeneous Landau equation. By interpreting the Landau operator as a "transport" operator with a properly regularized velocity field, we can define a particle solution. Moreover, the method can preserve critical physical properties, including the conservation of charge, momentum, and energy, as well as the decay of entropy. This new "characteristic" perspective of the Landau equation offers a natural particle framework to handle collisions and field effects in a unified manner and opens door to many exciting research directions. Specifically, we will pursue the following three aims, which, upon integration, will form a complete CPIC method for the full Vlasov-Landau-Maxwell (VLM) system: 1) Establishing proper particle regularization for the spatially dependent Landau collision operator. 2) Developing structure-preserving particle-field solver for the VLM system and validation it against plasma benchmarks. 3) Enhancing the efficiency and accuracy of the proposed CPIC method.

- **Please provide details of accomplishments during this reporting period**

  During the three-year period of this project, we have published 11 research papers in top journals of numerical analysis and scientific computing. Additionally, several papers have been submitted and are currently under review. Among these publications, we highlight the following accomplishments:

  - R. Bailo, J. Carrillo, and J. Hu. The collisional particle-in-cell method for the Vlasov-Maxwell-Landau equations. *J. Plasma Phys.*, 90:905900415, 2024.
  - J. Carrillo, J. Hu, and S. Van Fleet. A particle method for the multispecies Landau equation. *Acta Appl. Math.*, 194:6, 2024.
  - J. Hu, S. Van Fleet, and A. Wan. Fully discrete energy-dissipative and conservative discrete gradient particle methods for a class of continuity equations. Under minor revision, *SIAM J. Sci. Comput.*, 2024.

  In the first paper, we developed the CPIC method for the full VLM system. The method incorporates several key features to ensure accuracy and efficiency: it uses the random initialization to remove artifacts typical of deterministic initialization; it uses compactly supported B-spline as regularizers in both physical and velocity spaces to enhance efficiency; and it employs the Yee's lattice to ensure energy conservation. The method has been benchmarked on a series of 1D2V problems, including collisional Landau damping, two-stream instability, and Weibel instability.

  In the second paper, we extended the particle method to the multi-species Landau system, a critical step for realistic plasma simulations. A major contribution of this work was the discovery of a critical condition on the regularization parameters related to particle mass, ensuring the correct long-term behavior of the solution.

In the third paper, we proposed a novel time integrator based on discrete gradients. The resulting particle method can achieve conservation of mass, momentum, energy, and entropy decay at the fully discrete level. This is essential for obtaining accurate and stable simulations for long time.

- **How were the results disseminated to communities of interest?**

1) The submitted and published papers have been posted simultaneously on arXiv and the PI's personal webpage https://jingweihu-math.github.io/webpage/publication/ to enhance dissemination.
2) The PI participated in several online and in-person conferences, workshops, and seminars to present our research findings. These include U.S. National Congress on Computational Mechanics in 2023; the International Workshop on Moment Methods in Kinetic Theory at KIT, Karlsruhe, Germany in 2023; and seminar and colloquium talks at NYU Courant Institute of Mathematical Sciences in 2024, the PIMS Distinguished Colloquium at University of Northern British Columbia in 2023, among others. Additionally, the postdoc, Sam Van Fleet, also attended several conferences to present our research results, such as the SIAM Pacific Northwest Section Meeting in 2023 and SIAM Annual Meeting in 2024.

Impacts

- **Development of the principal discipline(s) of the project**

During this three-year project, we developed a comprehensive structure-preserving collisional particle-in-cell (CPIC) method for the full Vlasov-Landau-Maxwell system. These kinetic equations model particle transport, field effects, and particle collisions, and are regarded as first-principles physics for plasmas. With the proper inclusion of collisions, the newly developed CPIC method provides accurate simulations across a wide range of physical regimes for both low- and high-density plasmas. Furthermore, the method preserves critical physical properties, including the conservation of charge, momentum, and energy, as well as the decay of entropy. The preservation of these properties at the discrete level ensures faithful and robust numerical simulations, which is essential for understanding the complex behaviors of plasmas.

- **Describe the impact in this reporting period on the development of human resources**

The project provided summer support for the PI to work on the proposed topics. It also supported the postdoc, Sam Van Fleet, to work on the multi-species extension of the CPIC method and structure-preserving time integrator for the particle method. Additionally, it partially funded the PI and the postdoc to attend several conferences and workshops to present their research findings. In Summer 2024, the project also supported a beginning graduate student, Howard Cheng, in working on numerical methods for plasma simulations.

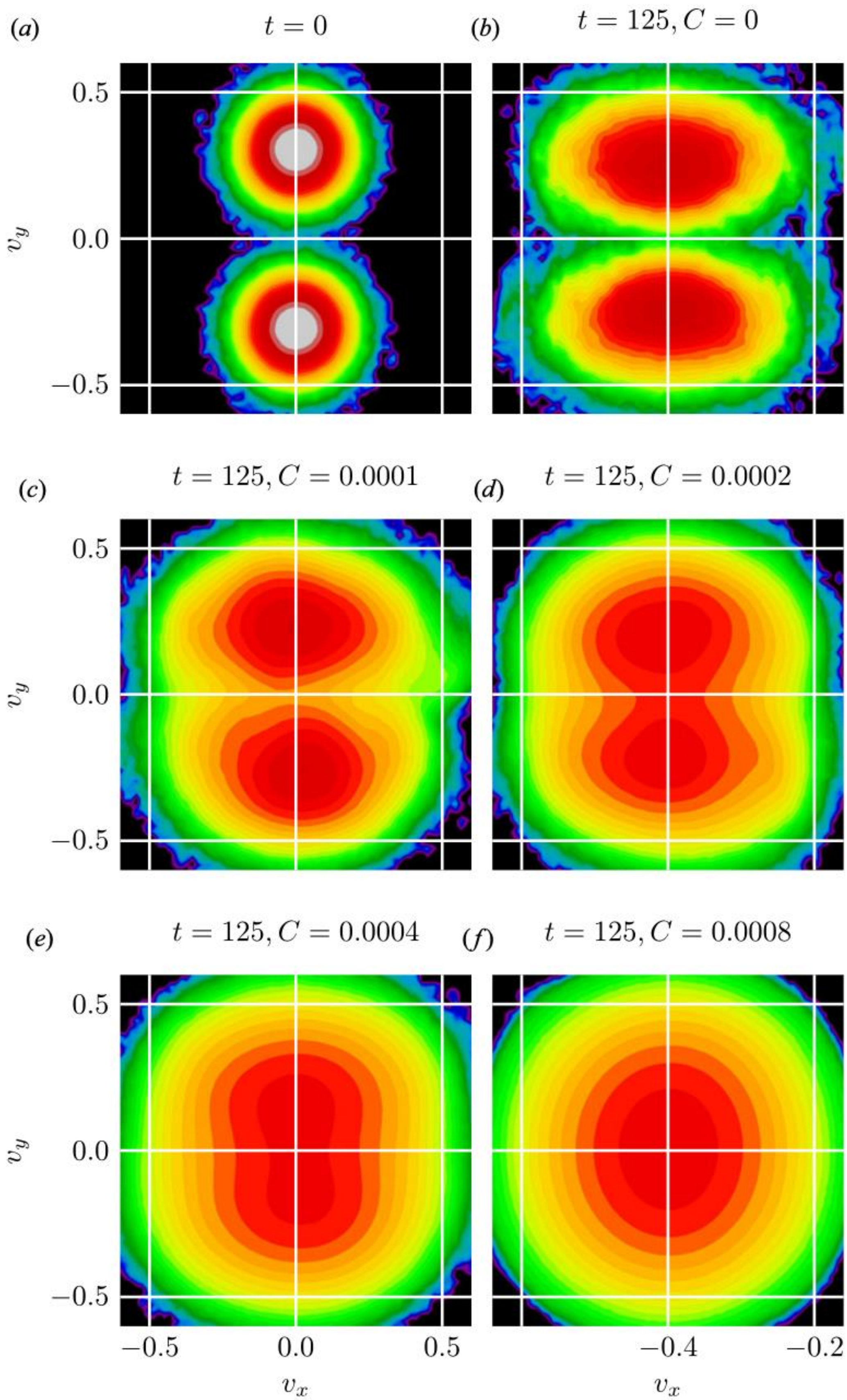
- **Impact on society beyond science and technology:**
  - Together with Fengyan Li and Sara Pollock, we are organizing a topical collection through *La Matematica* (the official journal of the Association for Women in Mathematics), featuring recent advances in numerical analysis and scientific computing. Submissions are currently open, and the PI is currently serving as the lead editor for this collection.
  - In Summer 2024, the PI served as a faculty mentor at the workshop "*Empowering a Diverse Computational Mathematics Community*" hosted by ICERM. This two-week long research and professional development workshop supported the retention and success of junior and mid-career computational mathematicians from underrepresented groups through research collaborations, networking and leadership development via peer-led learning communities.
  - The PI continues to serve as a faculty mentor for the *Math Alliance* program, which supports underrepresented groups in mathematics. This past year, her new mentee is Tisha Asuncion, a senior at the University of California Berkeley.

Changes

Not applicable.

Technical Updates

Please see below for the results of the CPIC method applied to the Weibel instability -- a classical plasma benchmark with varying collision strengths (gradually increasing from C=0 to C=0.0008). To the best of our knowledge, these are the first collisional results for the Weibel instability reported in the literature. Additional results, including videos, can be found here: https://rafaelbailo.com/papers/BCH_CPIC/.



(a)
$t = 0$
(b)
$t = 125, C = 0$
(c)
$t = 125, C = 0.0001$
(d)
$t = 125, C = 0.0002$
(e)
$t = 125, C = 0.0004$
(f)
$t = 125, C = 0.0008$
v_y
0.5
0.0
−0.5
−0.5
0.0
0.5
−0.4
−0.2
v_x